Image Matching Challenge 2025: Technical Whitepaper Scene Clustering and 3D Reconstruction using MAST3R

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Abstract

The Image Matching Challenge (IMC) 2025 presents a complex computer vision problem centered on "Doppelgangers"—visually similar but distinct scenes that confuse traditional Structure-from-Motion (SfM) pipelines. Our project leverages MAST3R, a deep learning framework that performs dense 3D reconstruction directly from image pairs using a Vision Transformer architecture. This whitepaper details our implementation, integrating MAST3R with a Sparse Global Alignment backend, enabling accurate scene clustering, 3D reconstruction, and outlier rejection in challenging urban and indoor scenarios.

1. Introduction

The 2025 Image Matching Challenge focuses on robust 3D reconstruction and scene clustering in the presence of visually similar ("doppelganger") scenes. Traditional SfM pipelines fail under repetitive structures and wide baselines, motivating the use of dense 3D matching with MAST3R.

2. Background and Related Work

2.1. Structure-from-Motion (SfM)

Traditional pipelines like COLMAP follow:

- Feature Extraction (SIFT, SuperPoint)
- Matching
- Geometric Verification (RANSAC)
- Triangulation & Bundle Adjustment

However, these fail for textureless, repetitive structures.

2.2. Deep Matching Models

Transformer-based matchers like LoFTR or SuperGlue improve wide-baseline matching, but still rely on 2D features. MAST3R predicts dense 3D points and confidence maps per pixel, directly providing geometric cues for reconstruction.

2.3. Code Snippet Example

Dense 3D matching can be seen in:

```
# From mast3r/retrieval/model.py
def forward_local(self, x):
   feat, attn = self.
       extract_features_and_attention(x)
   return how_select_local(feat, attn, self
       .nfeat)
```

3. MAST3R Architecture

AsymmetricMASt3R uses a Siamese ViT encoder with cross-attention decoder to produce:

- 3D Point Head: Predicts (x, y, z) per pixel.
- Confidence Head: Assigns reliability scores to matches.
- Descriptor Head: Local features for retrieval and reciprocal matching.

4. Full System Pipeline

1. Dataset Ingestion
2. Preprocessing (resizing, cropping)
3. Pairwise Image Matching
4. Graph Construction
5. Scene Clustering
6. Sparse Global Alignment (SparseGA)
7. Submission Generation

5. Figures: Qualitative Results

6. Technical Innovations

- Sparse Global Alignment (SparseGA) optimizes 3D points and camera poses.
- Robust Loss Functions (`gamma_loss`) for outlier tolerance.
- Low-Rank Depth Approximation (LoRA) reduces memory while enforcing consistency.

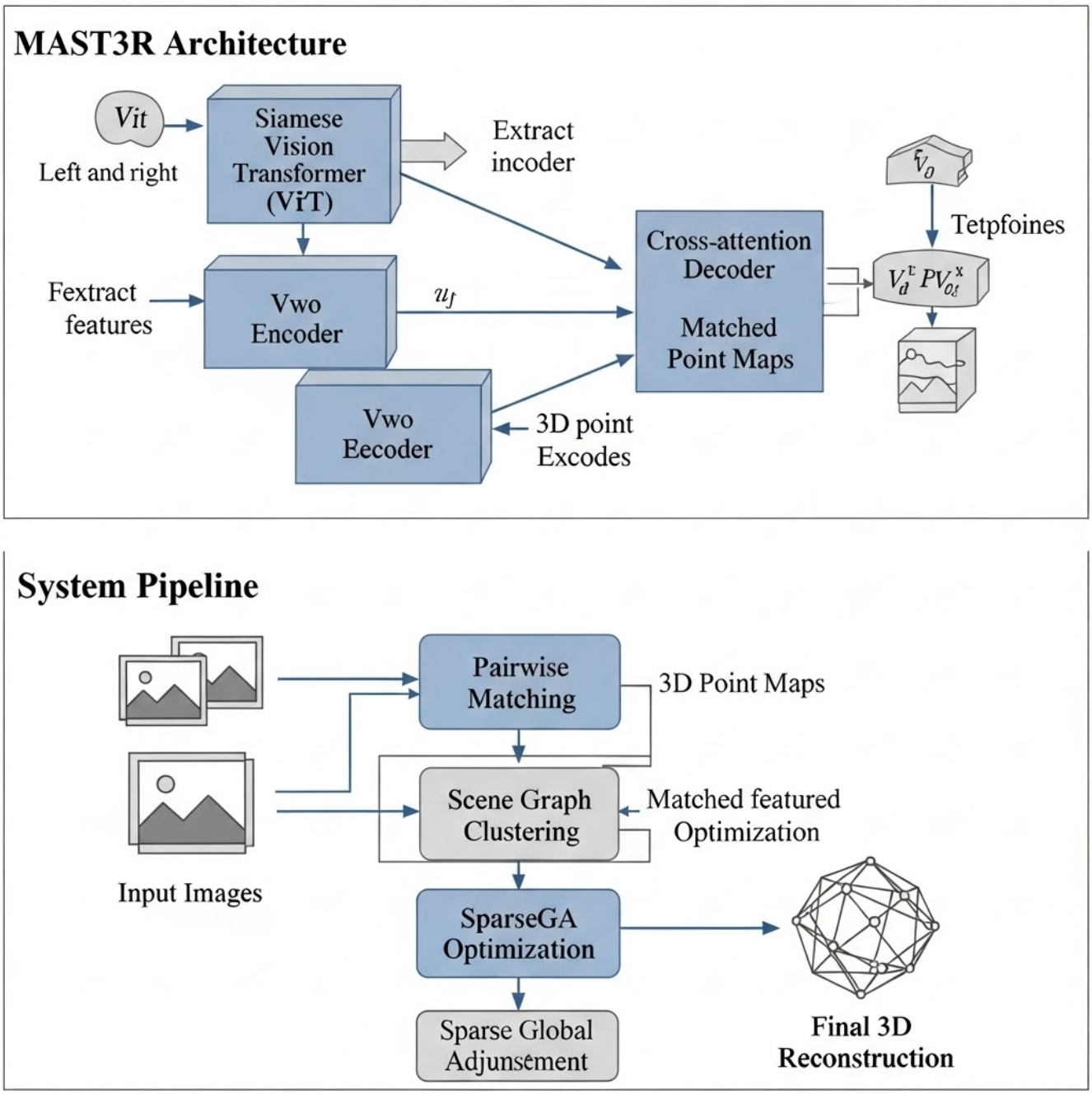


Figure 1. Project architecture and system pipeline.

7. Experiments & Results

Tested on IMC 2025 datasets:

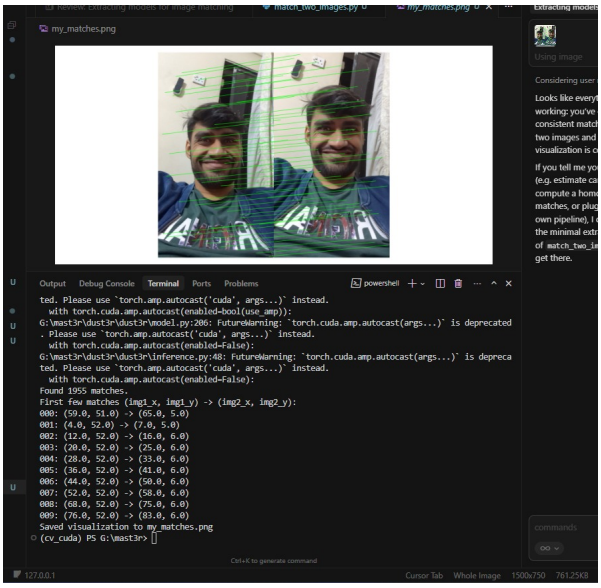
- Matching: Successful for less than 90° rotations.
- Clustering: Correctly separates doppelganger scenes.

8. Failure Cases & Observations

- Extreme repetitive structures can cause hallucinated matches.
- Disconnected components result in partial reconstructions.

(a) Geometric Consistency Check.

(b) Wide-Baseline Object Matching.



(c) Dense matching on faces.

Figure 2. Geometric consistency verification in an indoor scene with repetitive patterns **(a)** Dense semantic matching on facial features. **(b)** Handling of wide-baseline viewpoints on complex object geometry (bicycle). **(c)** Qualitative results from our MAST3R implementation demonstrating robustness across different scenarios.

9. Conclusion

MAST3R's dense 3D-first approach combined with SparseGA achieves robust clustering and reconstruction in challenging scenarios, setting a new standard for automated Structure-from-Motion systems.

10. References

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